Q 1.Consider a grammar G = (V , T , P , S) where-

- V = { S }
- T = { a , b }
- P = { S → aSbS , S → bSaS , S → ∈ }
- S = { S }

- This grammar is an example of a context free grammar.
- It generates the strings having equal number of a's and b's.

Q2. Language: of all strings having many 0's, defined over {0}

Regular Expression: a^+

Context Free Grammar: S ⇒ 0S | 0

Q.3

Language: of all strings having many 0's or no zero, defined over {0}

Regular Expression: a^*

Context Free Grammar: S ⇒ 0S | **ε**

Q4.

Language: of all strings having exactly one 0's or exactly one 1, defined over {0,1}

Regular Expression: 0+1

Context Free Grammar: S ⇒ 0 | **1**

Q 5. Language: of all strings having many 0's or many 1's or no zero or no one, defined over {0,1}

Regular Expression: (0+1)*

Context Free Grammar: S ⇒ 0S | **1**S | **ε**

5. Built the CFG for the language of all those strings having 0 a's or many a's?

S ⇒ aS | ε

Now we can read multiple a's from this CFG.

For example;

To read o a's from S ⇒ aS | ε

S ⇒ aS | ε

S⇒ ε

To read 1 a from S ⇒ aS | ε

S ⇒ aS

S⇒ a ε

S⇒ a

To read 2 a from S ⇒ aS | ε

S ⇒ aS

S ⇒ aaS

S⇒ a ε

S⇒ a

and similarly, we can read many a's from this CFG.

Q.7 Examples of CFG of odd Length strings

To read a;

CFG:

S → aX **|** bX
X→ aS | bS | ε

Path to Read the String

S ⇒ aX //Read first a.

X⇒ε //Read first aε.

finally, we can read it as a.

Congratulation, the string can be read and the machine can be stoped after successfully completing its task.

To read b;

CFG:

S → aX | bX
X→ aS | bS | ε

S ⇒ bX   //Read first b.

X⇒ε //Read first bε.

finally, we can read it as b.

Q. 6

CFG {N,T,P,S} be

N = {S}, T = {a, b}, Starting symbol = S, P = S → SS | aSb | ε

One derivation from the above CFG is “abaabb”

S → SS → aSbS → abS → abaSb → abaaSbb → abaabb

Q.7. Verify the the string aabbabba for the CFG given by is valid,

1. S → aB | bA
2. A → a | aS | bAA
3. B → b | bS | aBB

Q.8 Show the derivation tree for string "aabbbb" with the following grammar.

1. S → AB | ε
2. A → aB
3. B → Sb

Q. 9 Grammar: S → ASB | c A → ε | aA B → ε | bB

Derivation Sequences for acb

S => ASB => aASB => aSB => acB => acbB => acb

S => ASB => ASbB => ASb => aAcb => aAcb => acb

S => ASB => AcB => aAcB => aAcbB => acbB => acb

Problem-10:

Consider the grammar-

S → bB / aA

A → b / bS / aAA

B → a / aS / bBB

For the string w = bbaababa, find-

1. Leftmost derivation
2. Rightmost derivation

3. Parse Tree

Solution-

1. Leftmost Derivation-

S → b**B**

→ bb**B**B (Using B → bBB)

→ bba**B** (Using B → a)

→ bbaa**S** (Using B → aS)

→ bbaab**B** (Using S → bB)

→ bbaaba**S** (Using B → aS)

→ bbaabab**B** (Using S → bB)

→ bbaababa (Using B → a)

2. Rightmost Derivation-

S → b**B**

→ bbB**B** (Using B → bBB)

→ bbBa**S** (Using B → aS)

→ bbBab**B** (Using S → bB)

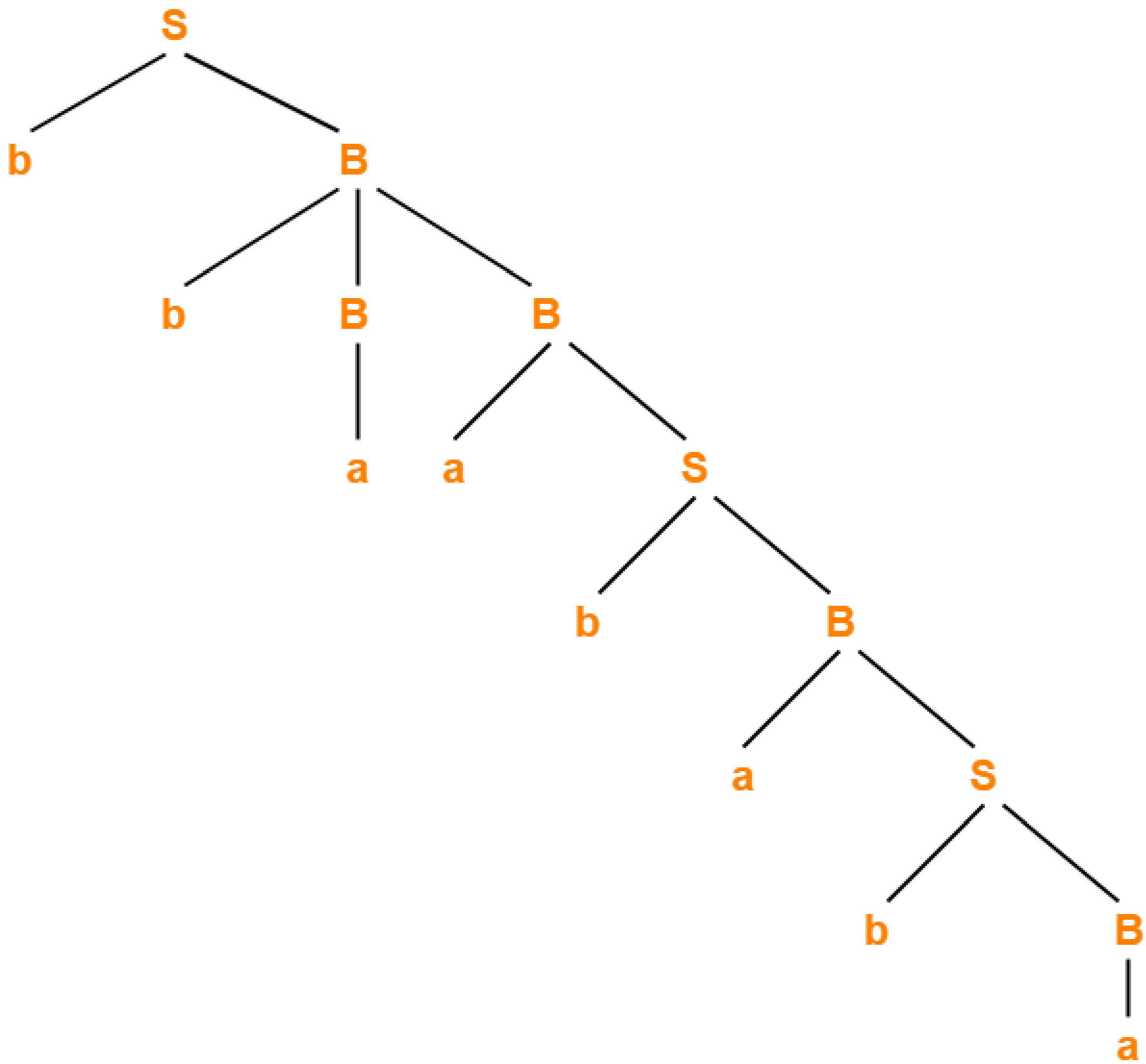
→ bbBaba**S** (Using B → aS)

→ bbBabab**B** (Using S → bB)

→ bb**B**ababa (Using B → a)

→ bbaababa (Using B → a)

3. Parse Tree-



Parse Tree

- Whether we consider the leftmost derivation or rightmost derivation, we get the above parse tree.
- The reason is given grammar is unambiguous.

Problem-11:

Consider the grammar-

$$S \to A1B$$

$$A \to 0A \mid \epsilon$$

$$B \to 0B \mid 1B \mid \epsilon$$

For the string w = 00101, find-

1. Leftmost derivation
2. Rightmost derivation
3. Parse Tree

Solution-

1. Leftmost Derivation-

S → **A**1B

→ 0**A**1B (Using A → 0A)

→ 00**A**1B (Using A → 0A)

→ 001**B** (Using A → ∈)

→ 0010**B** (Using B → 0B)

→ 00101**B** (Using B → 1B)

→ 00101 (Using B → ∈)

2. Rightmost Derivation-

S → A1**B**

→ A10**B** (Using B → 0B)

→ A101**B** (Using B → 1B)

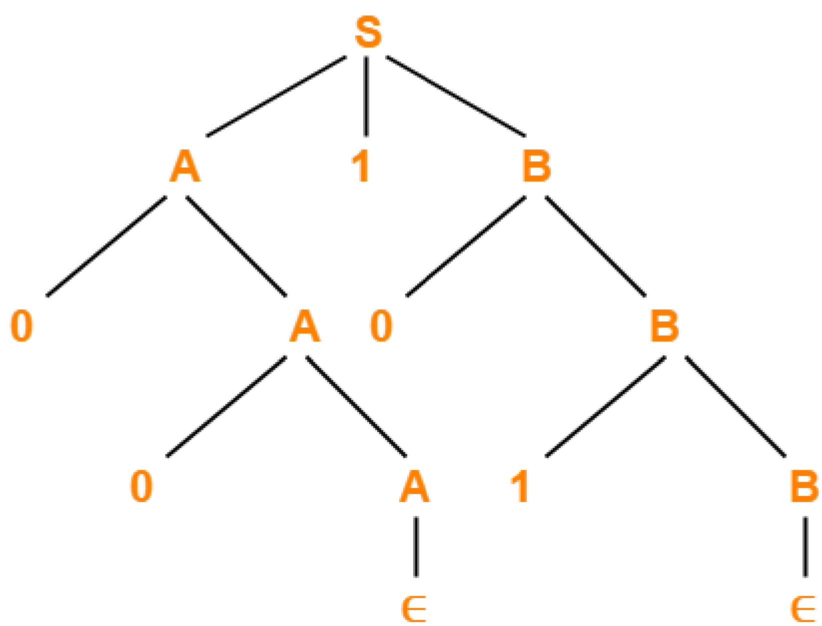
→ **A**101 (Using B → ∈)

→ 0**A**101 (Using A → 0A)

→ 00**A**101 (Using A → 0A)

→ 00101 (Using A → ∈)

3. Parse Tree-



Parse Tree

- Whether we consider the leftmost derivation or rightmost derivation, we get the above parse tree.
- The reason is given grammar is unambiguous.

Q. 12

Problem

Convert the following CFG into CNF

S → ASA | aB, A → B | S, B → b | ε

Solution

(1) Since **S** appears in R.H.S, we add a new state $\mathbf{S_0}$ and $\mathbf{S_0 \to S}$ is added to the production set and it becomes −

$S_0 \rightarrow S$, $S \rightarrow ASA \mid aB$, $A \rightarrow B \mid S$, $B \rightarrow b \mid \in$

(2) Now we will remove the null productions −

$B \rightarrow \in$ and $A \rightarrow \in$

After removing $B \rightarrow \varepsilon$, the production set becomes −

$S_0 \rightarrow S$, $S \rightarrow ASA \mid aB \mid a$, $A \rightarrow B \mid S \mid \in$, $B \rightarrow b$

After removing $A \rightarrow \in$, the production set becomes −

$S_0 \rightarrow S$, $S \rightarrow ASA \mid aB \mid a \mid AS \mid SA \mid S$, $A \rightarrow B \mid S$, $B \rightarrow b$

(3) Now we will remove the unit productions.

After removing $S \rightarrow S$, the production set becomes −

$S_0 \rightarrow S$, $S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$, $A \rightarrow B \mid S$, $B \rightarrow b$

After removing $S_0 \rightarrow S$, the production set becomes −

$S_0 \rightarrow ASA \mid aB \mid a \mid AS \mid SA$, $S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$

$A \rightarrow B \mid S$, $B \rightarrow b$

After removing $A \rightarrow B$, the production set becomes −

$S_0 \rightarrow ASA \mid aB \mid a \mid AS \mid SA$, $S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$

$A \rightarrow S \mid b$

$B \rightarrow b$

After removing $A \rightarrow S$, the production set becomes −

$S_0 \rightarrow ASA \mid aB \mid a \mid AS \mid SA$, $S \rightarrow ASA \mid aB \mid a \mid AS \mid SA$

$A \rightarrow b \mid ASA \mid aB \mid a \mid AS \mid SA$, $B \rightarrow b$

(4) Now we will find out more than two variables in the R.H.S

Here, $S_0 \rightarrow ASA$, $S \rightarrow ASA$, $A \rightarrow ASA$ violates two Non-terminals in R.H.S.

Hence we will apply step 4 and step 5 to get the following final production set which is in CNF −

$S_0 \rightarrow AX \mid aB \mid a \mid AS \mid SA$

$S \rightarrow AX \mid aB \mid a \mid AS \mid SA$

$A \rightarrow b \mid AX \mid aB \mid a \mid AS \mid SA$

$B \rightarrow b$

$X \rightarrow SA$

(5) We have to change the productions $S_0 \rightarrow aB$, $S \rightarrow aB$, $A \rightarrow aB$

And the final production set becomes −

$S_0 \rightarrow$ AX | YB | a | AS | SA

S→ AX | YB | a | AS | SA

A → b A → b |AX | YB | a | AS | SA

B → b

X → SA

Y → a

Example-13 – Let us take an example to convert CFG to CNF. Consider the given grammar G1:

```
S → ASB
A → aAS|a|ε
B → SbS|A|bb
```

Step 1. As start symbol S appears on the RHS, we will create a new production rule S0->S. Therefore, the grammar will become:

```
S0->S
S → ASB
A → aAS|a|ε
B → SbS|A|bb
```

Step 2. As grammar contains null production A-> ε, its removal from the grammar yields:

```
S0->S
S → ASB|SB
A → aAS|aS|a
B → SbS| A|ε|bb
```

Now, it creates null production B→ ε, its removal from the grammar yields:

```
S0->S
S → AS|ASB| SB| S
A → aAS|aS|a
B → SbS| A|bb
```

Now, it creates unit production B->A, its removal from the grammar yields:

```
S0->S
S → AS|ASB| SB| S
```

A → aAS|aS|a

B → SbS|bb|aAS|aS|a

Also, removal of unit production S0->S from grammar yields:

S0-> AS|ASB| SB| S

S → AS|ASB| SB| S

A → aAS|aS|a

B → SbS|bb|aAS|aS|a

Also, removal of unit production S->S and S0->S from grammar yields:

S0-> AS|ASB| SB

S → AS|ASB| SB

A → aAS|aS|a

B → SbS|bb|aAS|aS|a

Step 3. In production rule A->aAS |aS and B-> SbS|aAS|aS, terminals a and b exist on RHS with non-terminates. Removing them from RHS:

S0-> AS|ASB| SB

S → AS|ASB| SB

A → XAS|XS|a

B → SYS|bb|XAS|XS|a

X →a

Y→b

Also, B->bb can't be part of CNF, removing it from grammar yields:

S0-> AS|ASB| SB

S → AS|ASB| SB

A → XAS|XS|a

B → SYS|VV|XAS|XS|a

X → a

Y → b

V → b

Step 4: In production rule S0->ASB, RHS has more than two symbols, removing it from grammar yields:

S0-> AS|PB| SB

S → AS|ASB| SB

A → XAS|XS|a

B → SYS|VV|XAS|XS|a

X → a

Y → b

V → b

P → AS

Similarly, S->ASB has more than two symbols, removing it from grammar yields:

S0-> AS|PB| SB

S → AS|QB| SB

A → XAS|XS|a

B → SYS|VV|XAS|XS|a

X → a

Y → b

V → b

P → AS

Q → AS

Similarly, A->XAS has more than two symbols, removing it from grammar yields:

S0-> AS|PB| SB

S → AS|QB| SB

A → RS|XS|a

B → SYS|VV|XAS|XS|a

X → a

Y → b

V → b

P → AS

Q → AS

R → XA

Similarly, B->SYS has more than two symbols, removing it from grammar yields:

```
S0 -> AS|PB| SB
S → AS|QB| SB
A → RS|XS|a
B → TS|VV|XAS|XS|a
X → a
Y → b
V → b
P → AS
Q → AS
R → XA
T → SY
```

Similarly, B->XAX has more than two symbols, removing it from grammar yields:

```
S0-> AS|PB| SB
S → AS|QB| SB
A → RS|XS|a
B → TS|VV|US|XS|a
X → a
Y → b
V → b
P → AS
Q → AS
R → XA
T → SY
U → XA
```

PRACTICE PROBLEMS BASED ON CHOMSKY NORMAL FORM-

Problem-01:

Convert the given grammar to CNF-

$$S \rightarrow aAD$$

$$A \rightarrow aB \,/\, bAB$$

$$B \rightarrow b$$

$$D \rightarrow d$$

Solution-

Step-01:

The given grammar is already completely reduced.

Step-02:

The productions already in chomsky normal form are-

$B \rightarrow b$ ………..(1)

$D \rightarrow d$ ………..(2)

These productions will remain as they are.

The productions not in chomsky normal form are-

$S \rightarrow aAD$ ………..(3)

$A \rightarrow aB \,/\, bAB$ ………..(4)

We will convert these productions in chomsky normal form.

Step-03:

Replace the terminal symbols a and b by new variables C_a and C_b.

This is done by introducing the following two new productions in the grammar-

$C_a \rightarrow a$ ………..(5)

$C_b \rightarrow b$ ………..(6)

Now, the productions (3) and (4) modifies to-

$S \rightarrow C_aAD$ ………..(7)

$A \rightarrow C_aB \,/\, C_bAB$ ………..(8)

Step-04:

Replace AD and AB by new variables C_{AD} and C_{AB} respectively.

This is done by introducing the following two new productions in the grammar-

$C_{AD} \rightarrow AD$ ………..(9)

$C_{AB} \rightarrow AB$ ………..(10)

Now, the productions (7) and (8) modifies to-

$S \rightarrow C_aC_{AD}$ ………..(11)

$A \rightarrow C_aB \,/\, C_bC_{AB}$ ………..(12)

Step-05:

From (1), (2), (5), (6), (9), (10), (11) and (12), the resultant grammar is-

$S \rightarrow C_aC_{AD}$

$A \rightarrow C_aB \,/\, C_bC_{AB}$

$B \rightarrow b$

$D \rightarrow d$

$C_a \rightarrow a$

$C_b \rightarrow b$

$C_{AD} \rightarrow AD$

$C_{AB} \rightarrow AB$

This grammar is in chomsky normal form.

Problem-02:

Convert the given grammar to CNF-

$$S \rightarrow 1A \ / \ 0B$$

$$A \rightarrow 1AA \ / \ 0S \ / \ 0$$

$$B \rightarrow 0BB \ / \ 1S \ / \ 1$$

Solution-

Step-01:

The given grammar is already completely reduced.

Step-02:

The productions already in chomsky normal form are-

$A \rightarrow 0$ ………..(1)

$B \rightarrow 1$ ………..(2)

These productions will remain as they are.

The productions not in chomsky normal form are-

S → 1A / 0B ………..(3)

A → 1AA / 0S ………..(4)

B → 0BB / 1S ………..(5)

We will convert these productions in chomsky normal form.

Step-03:

Replace the terminal symbols 0 and 1 by new variables C and D.

This is done by introducing the following two new productions in the grammar-

C → 0 ………..(6)

D → 1 ………..(7)

Now, the productions (3), (4) and (5) modifies to-

S → DA / CB ………..(8)

A → DAA / CS ………..(9)

B → CBB / DS ………..(10)

Step-04:

Out of (8), (9) and (10), the productions already in Chomsky Normal Form are-

S → DA / CB ………..(11)

A → CS ………..(12)

B → DS ………..(13)

These productions will remain as they are.

The productions not in chomsky normal form are-

A → DAA ………..(14)

B → CBB ………..(15)

We will convert these productions in Chomsky Normal Form.

Step-05:

Replace AA and BB by new variables E and F respectively.

This is done by introducing the following two new productions in the grammar-

E → AA ………..(16)

F → BB ………..(17)

Now, the productions (14) and (15) modifies to-

A → DE ………..(18)

B → CF ………..(19)

Step-06:

From (1), (2), (6), (7), (11), (12), (13), (16), (17), (18) and (19), the resultant grammar is-

S → DA / CB

A → CS / DE / 0

B → DS / CF / 1

C → 0

D → 1

E → AA

F → BB

This grammar is in chomsky normal form.

To gain better understanding about Chomsky Normal Form,